

CPKC

Instructions for Remote Control Operation

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1.0 General Requirements

1.1 Application, Definitions, and Abbreviations

- Employees connected with the operation of remote-control locomotive equipment must have a copy of this document accessible while on duty.
- This document may be modified via General Order to include location-specific instructions or other changes.
- All GCOR, ABTH, Safety, and Timetable Special Instructions remain in effect, except as modified herein.
- A Remote Control Zone is a portion of track(s) within definite limits designated in the timetable special instructions. Timetable special instructions will designate remote control operating areas and zones.
- Only qualified operators or students who have been trained in remote control operations may operate remote control equipment.
- In these instructions, a “direct transfer” occurs when an OCU is handed directly to a relieving remote control operator.
- The independent brake override will be referred to as the independent brake selector.
- The train brake selector will be referred to as the automatic brake selector.
- Abbreviations:
 - RCO – Remote Control Operator.
 - RCZ – Remote Control Zone.
 - OCU – Operator Control Unit.
 - LCU – Locomotive Control Unit.
 - RCL – Remote Control Locomotive
 - RCLS – Remote Control Locomotive System.
 - RSC – Reset Safety Control.
 - SCU – Setup Control Unit.

1.2 Requirements for Remote-Control Operations

- Each RCO must have an operative radio in their possession.
- Unless otherwise provided for, RCOs must always wear their OCU when performing remote control duties.
- RCOs are responsible to ensure that they are familiar with current information regarding remote control systems and locomotives.
- An RCO shall control only one remote control consist at a time and shall not operate any other RCL or manual mode locomotive simultaneously.
- When attaching an OCU the four-point harness must be used. A separate clip shall be used for each of the four securement rings located on the OCU.
- A tag indicating that a locomotive is operating in remote control mode must be placed at the throttle of the control stand of each remote-control locomotive. The tag must be removed when the locomotive is returned to manual mode.
- Prior to beginning operations with remote control equipment, a job briefing must be held among all crew members. All remote control crew members must be informed and clearly understand which crew member will be controlling the movement with the OCU, and what tasks are to be performed.

- When in control of a remote control movement, do not transfer control to the other RCO until ready for movement.
- Operators must be vigilant to the approximate number of cars and tonnage of the equipment being handled to ensure compliance with proper train handling practices. Whenever there is doubt that locomotive brakes can control and stop a movement, sufficient air brakes must be cut-in to ensure adequate braking ability exists.
- The OCU must only be operated from a position on the ground, or from the movement being controlled. It is prohibited to operate the OCU from within a motorized vehicle. Except for main track movements, the OCU must not be operated within the cab of a locomotive.
- Immediately after commanding direction and speed, the controlling employee must visually verify that the movement is responding in the requested direction.
- Defective OCU's must be removed from service, submitted to the appropriate manager on duty, and tagged with the following information: Date / Time, name of reporting person, nature of defect, serial number of OCU.
- Strobe light(s):
 - Are located on the roof of the locomotive and will flash when the locomotive is operating in remote mode.
 - Must be operative when taking charge of RCLS equipment at an initial terminal.
 - At least one strobe light must be operative when taking charge of RCLS equipment at other than an initial terminal.
 - If one strobe light fails may continue without restriction. The failure must be reported to an on-duty supervisor.
 - If all strobe lights fail, a crew member must take up a position on the leading end of the working consist whenever that consist is in motion. The failure must be reported to an on-duty supervisor.
- Using RCLS equipment for a "running switch" (drop) is prohibited.
- The transmission of a "Low brake pipe pressure" or "Low main reservoir pressure" message indicates there is insufficient air and will prevent the movement from proceeding until sufficient pressure is detected. Whenever the state of air pressure is in doubt, stop the movement and recharge.
- RCLs and OCUs must not be left unattended while set up and communicating in remote control mode.
- Air brake tests can be performed with remote control locomotives, provided the following conditions are met:
 - Pressure and air flow requirements for the test being performed can be monitored and confirmed by either the OCU or the locomotive gauges.
 - The automatic brake reduction settings of the remote control system comply with Air Brake and Train Handling requirements for the test being performed.
- The speed selector must be placed in the Couple position no less than one car length from a coupling. Where possible, couplings should be made at 1 MPH or less.
- Brakes must be kept free of ice and snow in winter conditions by performing the following:
 - Condition the brake shoes periodically on locomotives by requesting a slower speed.
 - Move the independent brake selector to Low while in motion.
 - When in motion and operating with cars charged with air, toggle the automatic brake selector to the Min position.
 - Utilize the Snow Brake Feature, when equipped.

2.0 Taking Charge of RCLS Equipment

2.1 General

These instructions apply when taking charge of an RCLS consist left in remote mode or when initially setting up from manual to remote. These instructions do not apply when performing a direct transfer between crews.

When taking charge of RCLS equipment:

- Ensure a hand brake is applied.
- Verify that the flagging kit is fully supplied.
- Ensure the locomotive has received a locomotive daily inspection.
- Ensure locomotives are properly conditioned for remote control operation.

2.2 Control Stand Set Up

For Cattron systems, ensure the control stand of the remote locomotive is set up as follows:

- Headlights – Off.
- Throttle – Idle.
- Reverser Handle – Removed.
- Generator Field Switch – Off.
- Control/Fuel Pump and Engine Run – On.
- Automatic Brake – Cut In/Release.
- Independent Brake – Cut In/Release.
- RCL Man Down toggle switch – RCL Mode (when equipped).

For GE systems, ensure the control stand of the remote locomotive is set up as follows:

- Headlights – Off.
- Throttle – Idle.
- Reverser Handle – Removed.
- Generator Field Switch – Off.
- Control/Fuel Pump and Engine Run – On.
- Automatic Brake – Cut In/Release.
- Independent Brake – Cut In/Release.

2.3 Electrical Control Panel and Cabinet Set Up

Ensure the electrical control stand and cabinet of the remote locomotive is set up as follows:

- Alarm Silence Switch – On (if equipped).
- Number lights – On.
- Headlight – Conditioned per label.
- Isolation switch – Isolate.
- Battery knife switch – Fully closed.
- Circuit Breakers (black area) – On.

- Headlight Circuit Breaker – On.
- Remote Circuit Breaker – On.

2.4 RCL Set Up

Ensure RCL of the remote locomotive is set up as follows:

- Remote/Manual Selector Switch – Remote.
- Manual/Remote Transfer Valve – Remote, locking pin installed, when present.
 - For non-equipped locomotives with the SCU Relay Breaker, place the breaker to On.
- Isolation switch – Isolate.
- Validate strobe light(s) are functioning.

2.5 Programming Cattron OCUs

To program Cattron OCUs, perform the following:

- Ensure the OCU is turned off.
- Push the “NO” function button on the front panel of the LCU Interface Module for a minimum of 5 seconds. Follow the instruction on the LCU display screen.
- Line up the infrared eye on the LCU Interface Module with the infrared eye on the OCU.
- Press and hold the RSC/Vigilance button and turn “ON” the OCU. Listen to the chirping sound and follow the instruction on the LCU/OCU displays.
- When requested, select YES for linking SEC OCU option.
- Press and hold the RSC button and turn “ON” the OCU. Listen to the chirping sound and follow the instruction on the LCU/OCU displays.
- Recover emergency from the OCUs. The secondary OCU must be recovered first.
- When ready, release the automatic brake using the OCU.

2.6 Programming GE OCUs

To program GE OCUs, perform the following:

- Ensure the display shows GETS GS LOCOTROL RCL.
- Hold the OCU so the infrared port is pointed at the SCU infrared port.
- Press and release the PCC/LNK button on the OCU. The OCU will give a Linking LCU and Radio Testing message.
- Verify the OCU and the SCU are linked together by observing that the OCU displays Linked to Locomotive, LOCO ID: XXXX, and that the Green OCU #1 Linked status LED is lit. The first OCU that links to the SCU is defined as the Primary OCU and the Controlling OCU LED illuminates on the OCU to indicate that it is the controlling OCU for the RCL system.
- Once linked, the SCU will display Primary Tilt Test Not Complete.
- Press and hold the tilt override switch forward. Rotate the OCU greater than 45 degrees from horizontal until the SCU message disappears. The locomotive voice Radio will send a voice message that the tilt test is completed and the locomotive number.
- The RCL System remains in the Setup State (or transitions to the Setup State if it was in the Standby mode) with the Emergency Brakes applied, the Independent Brakes set to FULL, and the Throttle in IDLE, Release Independent Brakes on cabstand.

To link a secondary OCU, complete the following:

- Hold the secondary OCU so that the Infrared Port is pointed at the SCU Infrared port and press and release the PCC/LNK button on the OCU. The Primary OCU operator will get a Secondary OCU Link message on their display and within 5 seconds the operator must press the LR or RR button on the OCU to authorize the second OCU to Link.
- Verify that the second OCU linked by observing the Locomotive Road number on the secondary OCU display and that the Green OCU # 2 Linked status LED is lit. The Controlling OCU LED on the secondary OCU remains off.
- If the Setup Emergency had previously been recovered by the Primary OCU, the locomotive will be put into Park State. Once the secondary OCU is linked, a secondary OCU Tilt test must be performed using the same method as the primary Tilt test. If the Setup Emergency has not been recovered prior to linking a secondary OCU, the RCL System will remain in the Setup State.
- Upon successful completion of the Link operation, recovery of the Setup Emergency, and verifying the emergency valves, the Locomotive Control Unit (LCU) transitions into the Ready State. The Ready State allows an OCU operator to control the locomotive's speed and brakes. The SCU displays Remote Mode on the display.

3.0 Tests, Qualification, and Functions of Remote-Control Systems

3.1 Required Tests

When setting up the RCLS, the following tests must be performed:

- Status Test – Both OCUs. Must be completed first.
- Tilt Test – Both OCUs.
- Reset Safety Control (RSC) /Vigilance Test – Both OCUs.
- Miscellaneous Test – Both OCUs.
- Locomotive Brake Test with OCU – Controlling OCU.

3.2 Status Test

To ensure proper communication between the OCUs and locomotive, perform the following:

- Cattron
 - The primary OCU display indicates "Pri" for primary and "Sec" for secondary.
 - Toggle the OCU Status Button. Verify the locomotive number displayed is correct.
 - If communication fails with the OCU, the display will indicate "Poll Offline."
- GE
 - Ensure the CO LED is illuminated. This indicates the OCU is primary and is in control of the remote locomotive.
 - Toggle the Tilt Over-ride (TOR) button. Verify the locomotive number displayed is correct.

3.3 Tilt Test

A Tilt Test must be performed at least once during every 24-hour period of operation, at the start of each shift, and additionally when replacing a defective OCU, programming an OCU, or taking charge of an unattended consist.

To perform a Tilt Test, complete the following on both OCUs:

- Tilt the OCU more than 45 degrees for at least 5 seconds. An audible alarm will sound and the system will initiate an emergency brake application.
- Using the brake pipe pressure gauge, verify that the brake pipe pressure reduces to 0 psi.
- Verify the man down/tilt feature automatic notification message transmits over the radio.

- Recover the emergency from the OCU.
- Ensure the brakes recover.

3.4 Reset Safety Control (RSC) /Vigilance

The RSC/ Vigilance test must be performed each time the remote-control system is set up.

To test the RSC/Vigilance feature, perform the following:

- Ensure the hand brake is fully applied and the isolation switch is in the Isolate position.
- Set the speed selector position to Stop.
- Set the direction to Forward or Reverse.
- Press the RSC/Vigilance button.
- Within 10 seconds, set the speed selector to Coast.
- Wait 50 seconds for the RCL alarm to sound. The alarm will sound for 10 seconds followed by a full-service brake application.
- Verify the brakes apply.
- Recover from the full-service brake application when ready.

3.5 Miscellaneous Test

Miscellaneous testing must occur at the start of each shift, when replacing a defective OCU, programming an OCU, or taking charge of an unattended consist. To perform a miscellaneous test, check to ensure the bell, whistle, sanders, and headlights can be activated/controlled from the OCUs. When testing sanders, ensure sand is being deposited on the rail.

3.6 Locomotive Brake Test with OCU

A Locomotive Brake Test with OCU is required after a locomotive has been initially set up from manual to remote, when taking charge of RCLS equipment and a direct transfer has not occurred, and when adding a locomotive or setting out a locomotive from the middle of a consist.

To perform a locomotive brake test with OCU, complete the following:

- Ensure the hand brake(s) are released. The locomotive will move while performing the test.
- Ensure the automatic brake has been in Release for at least 2 minutes.
- Take position on the ground to observe the brake cylinders during the test.
- Set the reverser switch to Forward or Reverse.
- Press the RSC/Vigilance button and move the speed selector to Couple within 5 seconds.
- Ensure the locomotive(s) move in the selected direction and all brakes release.
- Set the independent brake selector to Low and ensure the brakes apply.
- Set the independent brake selector to Release and ensure the brakes release.
- Move the automatic brake selector to the Min position and ensure the brakes remain released.
- Move the automatic brake selector to the Full position and ensure the brakes apply and the locomotive stops.
- Place the speed selector in Stop.
- Place the automatic brake selector to Release.

3.7 Running Brake Test

A running brake test is required as follows:

- When taking charge of RCLS equipment.

- After powering on an OCU and/or resuming operations.

To perform a running brake test, complete the following:

- Ensure the hand brake(s) are released.
- Set the reverser direction to Forward or Reverse.
- Press the RSC/Vigilance button and move the speed selector to 4 MPH within 5 seconds.
- Set the independent brake selector to Low. Verify that brakes apply with sufficient force to slow the movement.
- Move the independent brake selector to Release.

3.8 Resuming Operations

After direct transfer of an OCU to a new RCO or anytime an OCU has been turned off, perform the following:

- If the OCU is off, turn on.
- Recover the brakes.
- Perform a Status Test.
- Perform a Tilt Test to audible alarm only.
- Perform a Running Brake Test.

3.9 Replacing an OCU

After replacing an OCU for defect or other reason, complete the following:

- Reprogram the OCU.
- Perform the following tests on the replacement OCU:
 - Status Test.
 - Tilt Test.
 - RSC /Vigilance Test.
 - Miscellaneous Test.
 - Locomotive Brake Test with OCU.

3.10 Recovery and Release Procedures

To recover from an emergency brake application, perform the following:

- Move the speed selector to Stop.
- Ensure the automatic brake selector is toggled to Release.
- Move independent brake selector to the emergency position. Wait for emergency to time out, if required.
- Press the RSC/Vigilance button and move the independent brake selector to the Release position.
- Verify that the train brakes release.

To recover a train brake penalty, perform the following:

- Move the speed selector to stop and place reverser in Neutral.
- Toggle the automatic brake selector to the full position/match the current locomotive state.
- Press the RSC/Vigilance button and within 5 seconds move the automatic brake selector to Release.
- Verify that the train brakes release.

To recover a service brake penalty, perform the following:

- Ensure the locomotive is stopped.
- Place the speed selector to Stop.

To release the train brakes, perform the following:

- Toggle the automatic brake selector to the Release position.
- Verify that the train brakes release.

3.11 Transfer of Control – Pitch and Catch Procedures

To transfer operator control using a Cattron system, perform the following:

- Ensure the locomotive is stopped.
- Set both OCU speed selectors to Stop and reversers to Neutral.
- Ensure both OCU automatic brake selector setting match.
- The controlling OCU presses the Pitch button.
- Within 10 seconds, the receiving RCO presses the RSC/Vigilance button.
- The new primary OCU will display “Pri” and the new secondary OCU will display “Sec.”
- If transfer fails, repeat the procedure.

To transfer operator control using a GE system, perform the following:

- Ensure the locomotive is stopped.
- The primary operator (A) requests a Pitch operation by pressing the PCC/LNK button on their OCU (if the Primary operator does not receive catch confirmation from the secondary operator within 10 seconds, the audible alarm sounds and a PITCH FAILED message is displayed. The primary OCU (A) remains in control of the LCU.).
- The Secondary operator (B) receives an audible signal and a Catch request on the OCU display. The Secondary operator (B) then presses the PCC/LNK button on the OCU to accept the catch. (if the Secondary operator does not receive catch confirmation from the primary operator within 10 seconds, the audible alarm sounds and a PITCH FAILED message is displayed. The Primary OCU (A) remains in control of the LCU.)
- The Primary operator (A) receives an audible signal and a Catcher Ready on the OCU display. The Primary operator (A) again presses the PCC actuator on the OCU to confirm the pitch.
- The LCU then reassigns the primary operator as B and the secondary operator as A. The primary operator will have its Controlling OCU LED on.
- The Automatic Brake settings and Headlight control settings are automatically transferred to the new Primary OCU when the pitch and catch is successfully completed.
- If transfer fails, repeat the procedure.

3.12 Single OCU Operation

If an OCU becomes inoperative during a shift equipped with two RCOs, it is permissible to continue work, but only to:

- Clear a main track or lead.
- Clear public crossing at grade or cut the train as required.
- Move locomotive(s) to a designated track.

- Complete spotting of a customer, if no switching is required. The controlling OCU must perform the spot.

3.13 Dismissing and Reconnecting an OCU

To dismiss an OCU, complete the following:

- Ensure the locomotive is stopped with the reverser in Neutral.
- The Secondary RCO must hold both RSC/Vigilance buttons for at least 5 seconds.
- The Primary RCO must confirm the request to dismiss the OCU by pressing the RSC/Vigilance button within 5 seconds.
- Turn off the Secondary OCU.

To reconnect a dismissed OCU, complete the following:

- Ensure the locomotive is stopped with the reverser in Neutral.
- Power on the Secondary OCU.
- Hold both RSC/Vigilance buttons of the primary OCU for 5 seconds.
- The Secondary RCO must confirm the request to reconnect the OCU by pressing the RSC/Vigilance button within 5 seconds.
- The system will initiate an emergency application.
- Complete steps for resuming operation.

3.14 Charging

To utilize the charging feature using a Cattron system, perform the following:

- Toggle the automatic brake selector to the charge position. Press the RSC/Vigilance button for one second.
- Toggle the automatic brake selector to Release.

To utilize the charging feature using a GE system, perform the following:

- Press the RSC/Vigilance button on the OCU and immediately move the automatic brake selector to the charge position. Alternatively, from the release position, pull back on the automatic brake selector for a minimum of 5 seconds.
- To release from charge, pull back on the automatic brake selector momentarily.

3.15 Pull Back Stop Protection Conditioning

Each time a remote control locomotive is placed in service and at the first practical time after the start of each shift (not to exceed 2 hours), locomotives using pull-back stop protection must perform a conditioning run over a track where the system is utilized to ensure the system is functioning as intended.

4.0 Providing Protection

4.1 3-Point Protection

When required, perform the following steps to apply 3-point protection with an OCU:

- Place the speed selector in Stop.
- Ensure the reverser is in Neutral.

- Place the automatic brake selector to Light or greater, if required.
- Communicate 3-point protection per safety and operating rules.

5.0 Removal of OCU from Harness

5.1 Before Removing an OCU

Prior to removing an OCU, ensure:

- The locomotive is stopped.
- Control has been transferred to the RCO removing the OCU.
- A Job briefing has been completed with other crew members communicating the intent to remove the OCU.
- 3-point protection has been applied.

5.2 Protecting OCUs

When an OCU is removed, it must be attended by the RCO or placed in a designated area where it is secured from unauthorized access and tampering.

5.3 Commencing Operation following Removal of OCU

Before commencing operation:

- The OCU must be reattached as required.
- A Tilt Test to audible alarm must be performed.
- Ensure a job briefing has been performed.
- If the automatic brake was applied, release the automatic brake.

6.0 Securing Equipment

6.1 Testing Equipment Securement with a Remote Control Locomotive

To perform a test securement using a remote control locomotive, perform the following:

- Apply hand brakes in accordance with securement rules.
- Ensure the automatic brake selector is in Release.
- Select a direction.
- Press the RSC/Vigilance button and move the speed selector to Couple.
- Ensure the brakes release.
- Move the speed selector to Coast.
- Verify the movement stops, then move the speed selector to Stop.
- Immediately press the RSC/Vigilance button and move the speed selector to Coast. Ensure the equipment remains stationary.
- Move the speed selector to Stop.

6.2 Leaving a Locomotive Unattended in Remote Mode

- Secure the equipment in accordance with securement rules.
- Test securement as required by Testing Equipment Securement with a Remote Control Locomotive.

- Comply with locomotive shut down requirements.
- Turn off both OCUs.
- Verify locomotive headlights are Off.
- Close windows and lock doors as required

6.3 Leaving a Locomotive Unattended in Conventional Mode

- Ensure the speed selector of each OCU is set to Stop.
- Secure the equipment in accordance with securement rules.
- Test securement as required by Testing Equipment Securement with a Remote Control Locomotive.
- Turn off each OCU.
- Place the isolation switch in Isolate.
- Move the Manual/Remote Transfer Valve to the manual position and reinstall the locking pin.
- Ensure the independent brake is fully applied.
- When equipped:
 - Turn the three-position cut out switch to Lead.
 - Move the Remote/Manual Selector switch to manual (GE).
 - Place the RCL Man down Toggle Switch in Manual Mode.
- Move the automatic brake handle from Handle Off to Emergency.
- After 60 seconds, move the automatic brake handle to Release.
- Ensure the PCS light goes out and brake pipe pressure increases.
- Once recovered, leave the automatic brake handle conditioned as required by Air Brake and Train Handling rules.
- Verify locomotive headlights are Off.
- Comply with locomotive shut down requirements.
- Close windows and lock doors as required.

7.0 Snow Brake Operation

7.1 Use Requirements

The Snow Brake feature must be used when a locomotive is equipped, and:

- Ambient temperature is at or below 14° F.
- Required by local instruction or company officer.

The Snow Brake feature may be used at the discretion of the RCO when:

- Ambient temperature is above 14° F.

To prevent unnecessary brake shoe wear, unless operating conditions warrant, the snow brake feature must not be used when the ambient temperature is at or above 32° F.

7.2 Activation of Snow Brake

To activate the snow brake, perform the following:

- Press the MENU/NO button on the LCU Display Unit and hold it for 5 seconds will get the RCL Software to prompt the operator with the Apply Snow Brake? Message.
- Press ALARM RESET/YES button.
- A Snow Brake Activated message will appear.

7.3 Deactivation of Snow Brake

To deactivate the Snow Brake, perform the following:

- Press the MENU/NO button on the LCU Display Unit and hold it for 5 seconds will get the RCL Software to prompt the operator with the Release Snow Brake? Message.
- Press the ALARM RESET/YES button.
- A Snow Brake Deactivated message will appear.

7.4 Snow Brake Status

The Snow Brake Status can only be seen at the RCL Remote Display and is not available on the OCU display. The operator must press the MENU/NO button and hold it for 5 seconds to view status.

The RCL System will display the either of the following:

- Apply Snow Brake? Press Yes or No –This indicates that the snow brake status is not active.
- Release Snow Brake? Press Yes or No – This indicates the snow brake status is active.